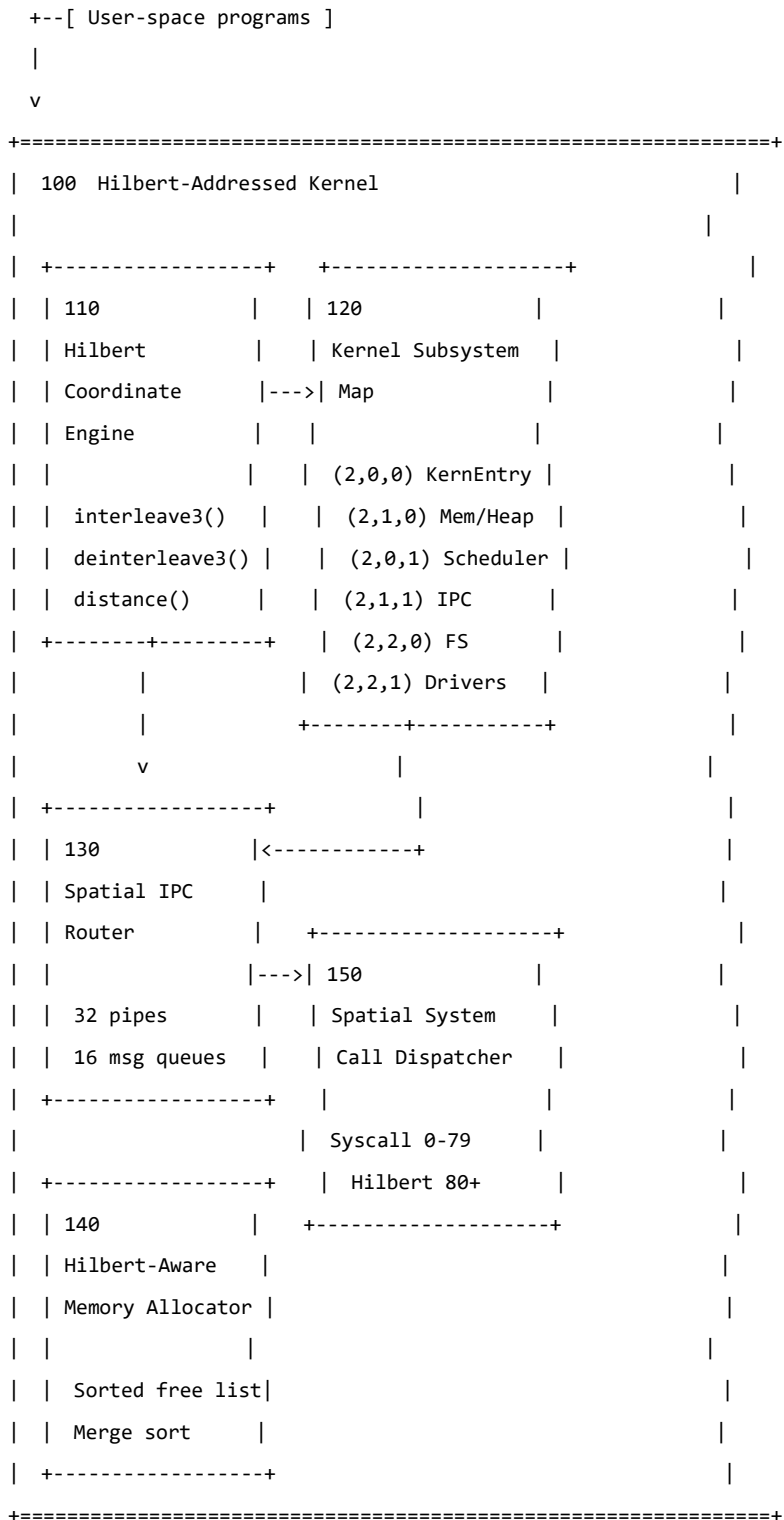


APPLICATION #1: HILBERT CURVE OS ARCHITECTURE

22 Patent Drawings -- ASCII Reference Diagrams

for tracing with Draw.io / Visio / PowerPoint

FIG. 1 -- System Architecture Block Diagram



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|
v
+--[ Physical hardware (x86_64) ]

```

FIG. 2 -- 3D Hilbert Coordinate Assignment

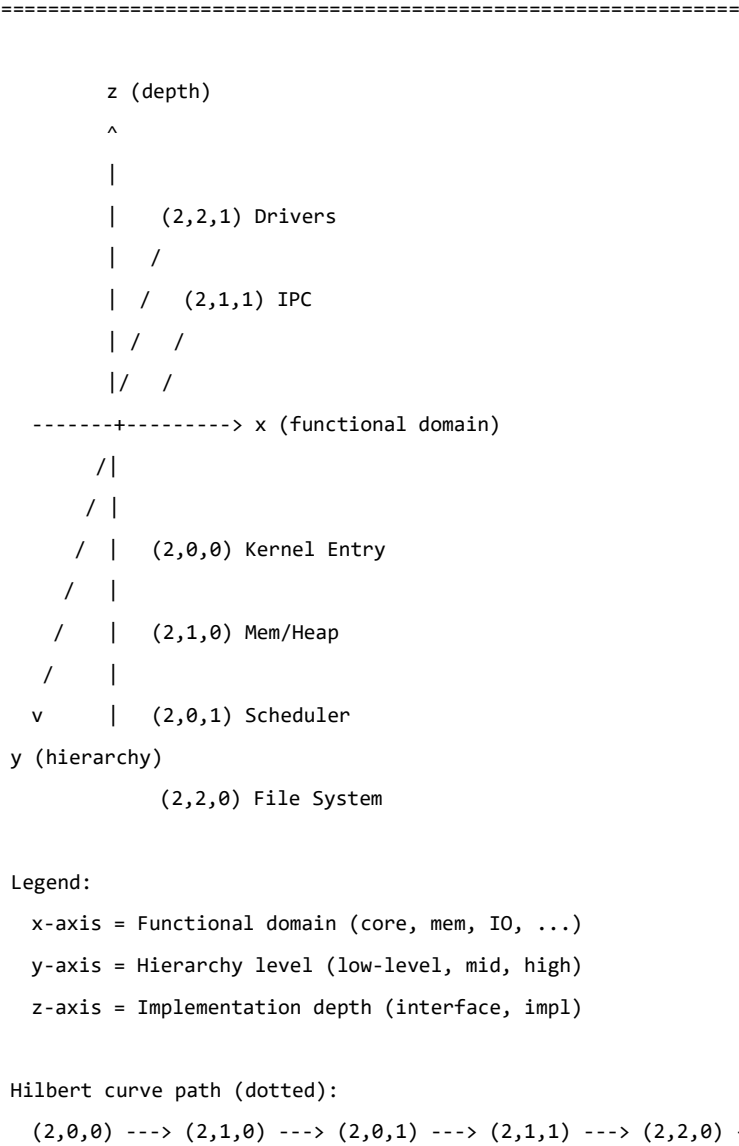


FIG. 3 -- Bit-Interleaving Encoding Flowchart (interleave3)



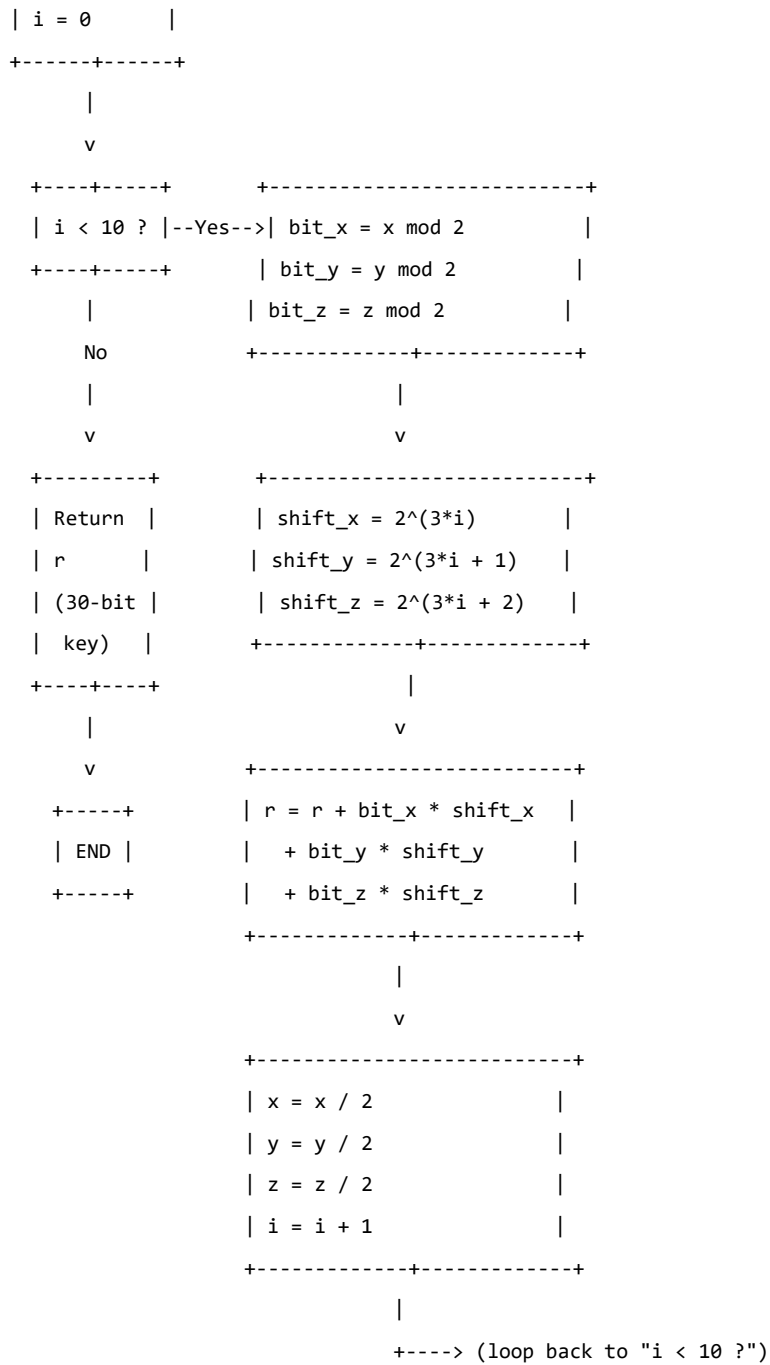
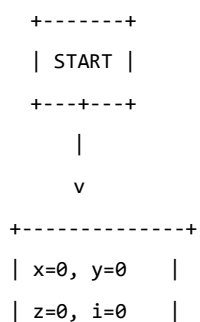


FIG. 4 -- Reverse Decoding Flowchart (deinterleave3)

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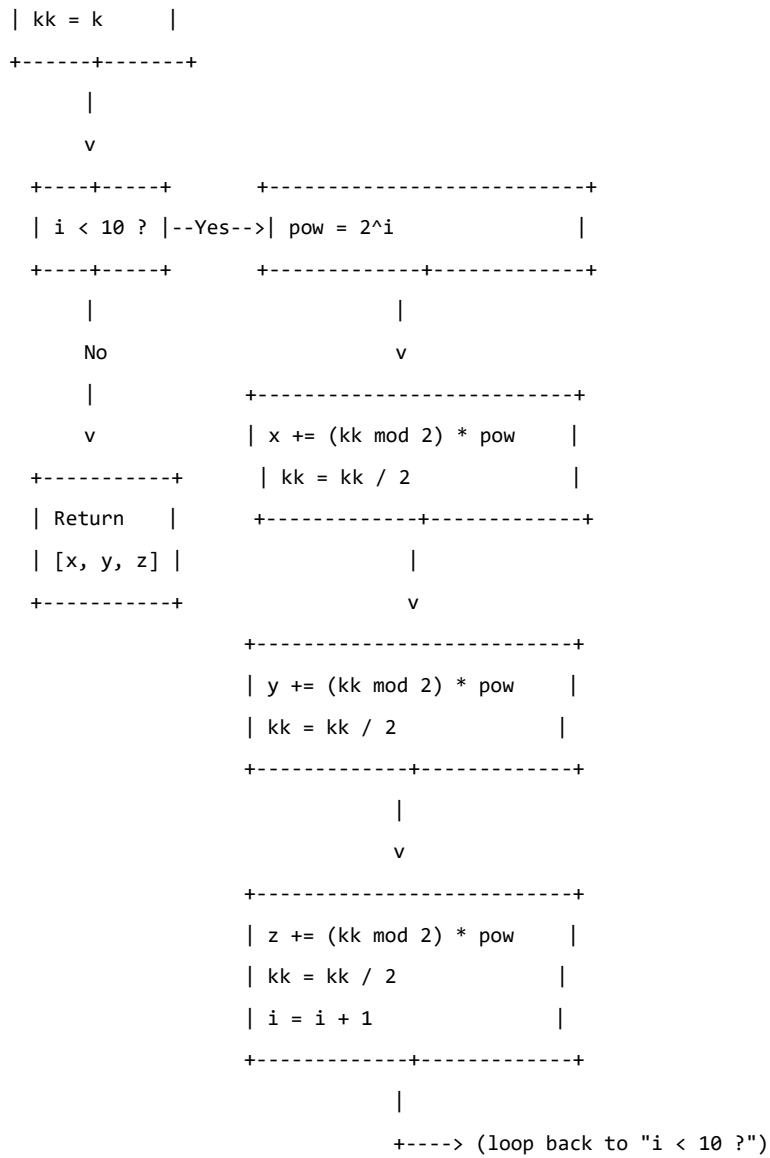


FIG. 5 -- 2D Hilbert Curve for Page Allocation (8x8)

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Column:	0	1	2	3	4	5	6	7
Row 0	0	1	14	15	16	19	20	21
Row 1	3	2	13	12	17	18	23	22
Row 2	4	7	8	11	30	29	24	25
Row 3	5	6	9	10	31	28	27	26
Row 4	58	57	54	53	32	35	36	37
Row 5	59	56	55	52	33	34	39	38

	+-----+-----+-----+-----+-----+-----+-----+-----+
Row 6	60 61 50 51 46 45 40 41
	+-----+-----+-----+-----+-----+-----+-----+-----+
Row 7	63 62 49 48 47 44 43 42
	+-----+-----+-----+-----+-----+-----+-----+-----+

Arrow: Curve traversal 0 -> 1 -> 2 -> 3 -> 4 -> ... -> 63

Note: "N must be power of 2"

Highlighted: nearby 1D indices (e.g., 10,11,12,13) occupy nearby 2D cells

FIG. 6 -- IPC Spatial Routing Mechanism

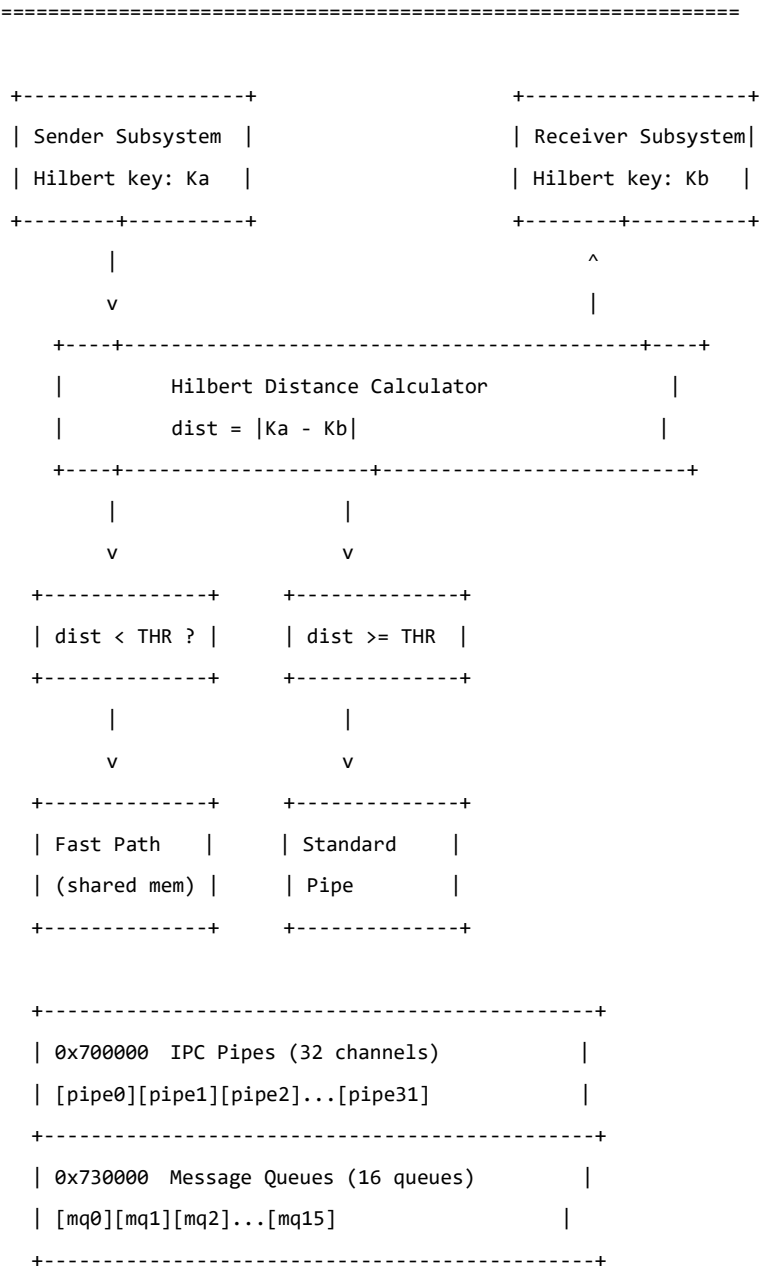
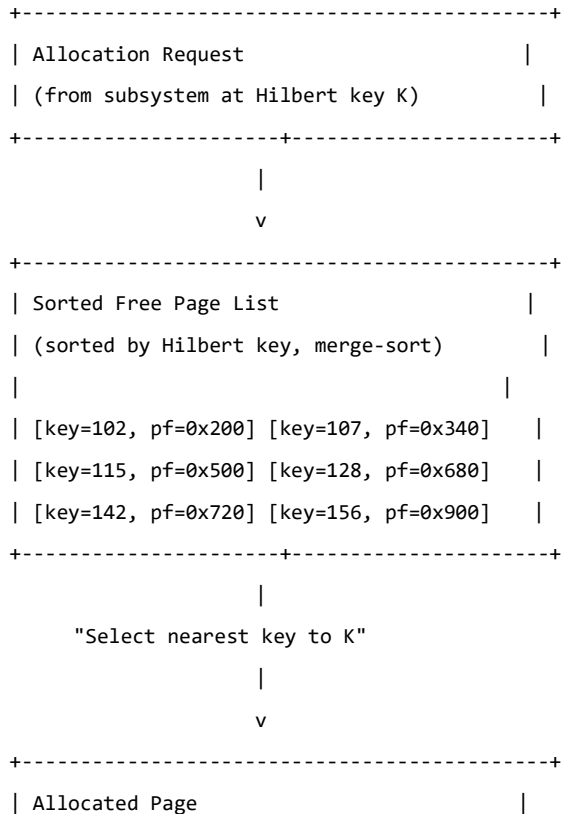


FIG. 7 -- System Call Dispatch by Hilbert Layout

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```

+-----+-----+-----+			
Syscall	Functional	Hilbert Region	
Range	Domain	Coordinates	
+-----+-----+-----+			
0 - 15	Process Control	(2,0,0)-(2,0,3)	
+-----+-----+-----+			
16 - 31	Memory Management	(2,1,0)-(2,1,3)	
+-----+-----+-----+			
32 - 47	File System	(2,2,0)-(2,2,3)	
+-----+-----+-----+			
48 - 63	Device I/O	(2,3,0)-(2,3,3)	
+-----+-----+-----+			
64 - 79	Network	(2,4,0)-(2,4,3)	
+-----+-----+-----+			
>= 80	** Hilbert Ops **	(2,5,0)+	
	80=ENCODE		
	81=DECODE		
	82=DISTANCE		
+-----+-----+-----+			

FIG. 8 -- Hilbert-Aware Memory Allocator



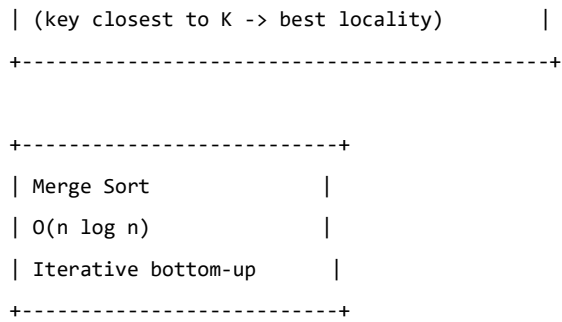


FIG. 9 -- Boot Chain Architecture

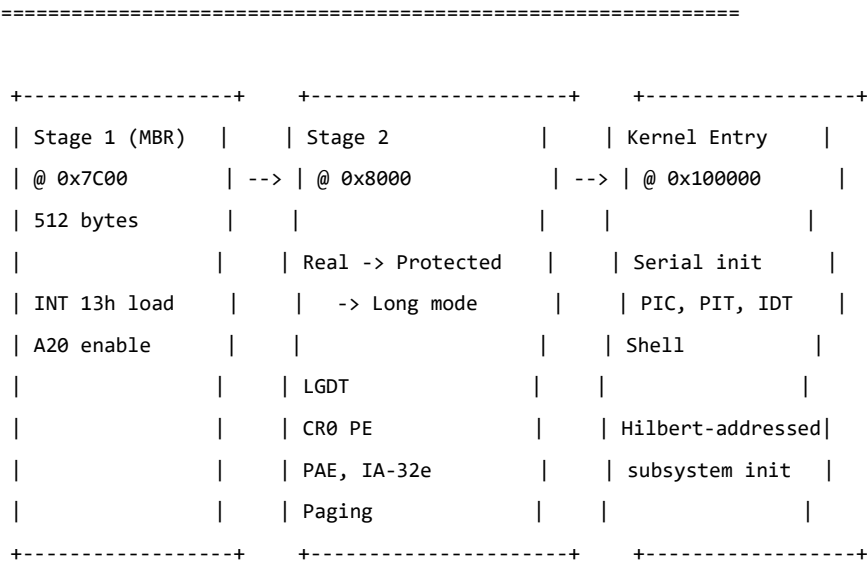


FIG. 10 -- Kernel Memory Map with Hilbert Annotations

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Address Range	Content	Hilbert
0x001000 - 0x003FFF	Page Tables (PML4/PDPT/PD)	
0x007C00 - 0x007DFF	Stage 1 (MBR)	
0x008000 - 0x00BFFF	Stage 2	
0x070000	Stack Top (grows down)	
~~~~~		
0x100000 - 0x1FFFFFF	Kernel Code (1 MB)	(2,0,0)
0x200000 - 0x200FFF	IDT (4 KB)	
0x300000 - 0x300017	Tick/Scancode/Key Counters	
~~~~~		
0x700000 - 0x720FFF	IPC Pipes (32 channels)	(2,1,1)
0x730000 - 0x770FFF	Msg Queues (16 queues)	(2,1,1)
~~~~~		
0x900000 - 0x901FFF	Module Table (64 slots)	
0x902000 - 0x902FFF	Trampoline Table (256)	

0x903000 - 0x942FFF      Module Code Arena (256 KB)

```
+-----+
| 0x000000 |
| [Page Tables ] |
| [MBR      ] 0x7C00 |
| [Stage 2   ] 0x8000 |
| [Stack     ] 0x70000 |
| ~~~~~~ |
| [Kernel Code      ] 0x100000 <-- (2,0,0) |
| [IDT             ] 0x200000 |
| [Counters        ] 0x300000 |
| ~~~~~~ |
| [IPC Pipes         ] 0x700000 <-- (2,1,1) |
| [Msg Queues        ] 0x730000 <-- (2,1,1) |
| ~~~~~~ |
| [Module Table      ] 0x900000 |
| [Trampoline        ] 0x902000 |
| [Module Arena      ] 0x903000 |
+-----+
```